

REPORT

Helix OTA — RK3588 / Orange Pi 5 Max Dev-Host A/B Emulator — Feasibility Research

Field	Value
Revision	1
Last modified	2026-06-11T08:30:00Z
Status	active — research (no implementation; one recommended first slice)
Status summary	Deep, cited (§11.4.8 / §11.4.99 / §11.4.123) feasibility study for a DEV- HOST-RUNNABLE RK3588 / Orange Pi 5 Max device emulator exercising a real OTA A/B apply→slot- switch→verify→auto-rollback mechanism WITHOUT physical hardware, as an extension of the containers submodule (§11.4.76). Honest fidelity ladder; recommended first buildable+verifiable slice; §11.4.74 reuse decision. Brutally honest about what cannot run on this Mac (§11.4.6 / §11.4.112).
Authority	Operator mandate 2026-06-10
Related	../emulation_infra/ REPORT.md; ../../design/ emulation_fabric/ ROADMAP.md; ../../design/ EMULATED_DEVICE_TESTING.md; ../../ design/CUTTLEFISH_TIER2.md; containers/ pkg/vm,pkg/boot,pkg/ compose,pkg/health; submodules/ota- update-engine-bridge, ota-android- agent, ota-protocol

Research discipline (§11.4.6/§11.4.123): every load-bearing claim cites an authoritative URL (accessed 2026-06-11). Unestablished facts are **UNCONFIRMED**:. Host-impossible items are §11.4.112. Effort figures are flagged estimates, never asserted as fact.

0. Host fact framing everything (FACT, §11.4.6)

Verified on this host 2026-06-11: Apple M3 Pro, macOS 15.5 (Darwin 24.5.0), arm64; kern.hv_support: 1 (HVF), NO /dev/kvm; M3 lacks hardware nested-virt (M4+/macOS-15 only); QEMU 11.0.1; podman 5.8.2; pandoc+weasyprint present.

Consequences: (1) any KVM-gated Android virtualization (Cuttlefish/redroid/K8s-STF) is host-gated, not impossible; (2) the only accelerated path is QEMU/AVD running a native arm64 guest on HVF — an **arm64 Linux guest under QEMU+HVF runs near-native** (the lever the recommendation rests on).

1. Q1 — QEMU RK3588 board model? (FACT)

No. There is no RK3588/RK35xx machine model in upstream QEMU. Realistic aarch64 target = generic -machine virt (already what containers/pkg/vm/qemu.go boots). Confirmed: no Rockchip SoC QEMU machine; guidance is generic virt, which “won’t be an rk3328 emulation, too many discrepancies” (OSDev.org “Emulating a RK3328 SoC?”, <https://forum.osdev.org/viewtopic.php?f=1&t=37324> ; qemu/u-boot rockchip Kconfig <https://github.com/qemu/u-boot/blob/master/arch/arm/mach-rockchip/Kconfig>).

Fidelity of virt vs real RK3588: virt provides a REAL bootloader (mainline U-Boot qemu_arm64 or edk2 UEFI) + REAL GPT slots on virtio-blk + REAL kernel + REAL dm-verity → the entire OTA apply/slot/verify/rollback state machine is exercisable at full fidelity. It does NOT model the RK3588 silicon (no Mali-G610, no VPU, no Rockchip PMIC/clock tree, no vendor TPL/SPL/idbloader boot chain). Silicon fidelity = real-hardware (T3) only.

2. Q2 — Real Android update_engine A/B on this Mac by ANY means? (cited)

No clean evidence-producing path on this M3 today. - Cuttlefish: requires Linux + /dev/kvm (AOSP cuttlefish-use <https://source.android.com/docs/setup/create/cuttlefish-use>). Host-gated §11.4.112 — runs on Linux+KVM/M4+/GCE-nested-virt. Not impossible. - Stock AVD (arm64-v8a HVF): runs real Android userspace, NOT a real update_engine/A/B-slot/AVB env. Whether a custom aosp_arm64_ab GSI + update_device.py applies a REAL payload under stock AVD with capturable slot-switch evidence is **UNCONFIRMED**: (GSIs exist + built from emulator images — AOSP GSI <https://source.android.com/docs/core/tests/vts/gsi> — but no source establishes stock AVD exposes a working update_engine A/B apply). Per §11.4.123 this is a research-trigger (GSI-A/B empirical test), NOT a bluff licence; claiming it works today would be a §11.4 bluff. - redroid: no update_engine/bootloader at all.

§11.4.112 classification (FACT): Real Android update_engine A/B on THIS M3 Mac is host-gated (Cuttlefish on Linux+KVM is canonical) + the AVD-GSI alternative is UNCONFIRMED — NOT structurally impossible. So the dev-host-now deliverable CANNOT be “real Android update_engine”; it MUST be the genuine-but-different A/B mechanism in §3.

3. Q3 — Dev-host-feasible alternative: real A/B on QEMU virt+HVF Linux (FACT + reuse)

A QEMU aarch64-virt Linux guest (HVF-accelerated) with: (1) a REAL GPT 2-slot (A/B) layout on virtio-blk; (2) a REAL bootloader (mainline U-Boot qemu_arm64) doing slot selection + AUTO-ROLLBACK via bootcount/bootlimit/altbootcmd/upgrade_available; (3) dm-verity per rootfs slot. An OTA agent writes the payload to the INACTIVE slot, flips active + arms the boot counter; the system reboots into the new slot, verifies dm-verity, marks good; a deliberately-CORRUPTED slot fails to boot and U-Boot AUTO-ROLLS-BACK.

Why it’s a real mechanism, not a mock (FACT): the U-Boot bootcount auto-rollback is the SAME engine real RK3588/embedded U-Boot uses — “if bootcount exceeds bootlimit, altbootcmd runs ... for A/B, altbootcmd swaps the rootfs partition ... a userspace app resets bootcount/upgrade_available to 0 on successful boot” (U-Boot Boot Count Limit <https://docs.u-boot.org/en/latest/api/bootcount.html> ; Mender U-Boot integration <https://docs.mender.io/operating-system-updates-yocto-project/board-integration/bootloader-support/u-boot/manual-u-boot-integration>). dm-verity is the identical kernel feature. The slot disk is a real GPT — only the block transport (virtio-blk vs eMMC) + SoC packaging differ.

Proves (REAL, on this Mac): payload→inactive slot; active-slot flip in real bootloader env; reboot into new slot; dm-verity verify; auto-rollback on corrupt slot via real bootcount/altbootcmd; ota-protocol round-trip + agent verify-before-apply driving a real apply. **Does NOT prove:** real Android update_engine/AVB; RK3588 vendor U-Boot packaging; Mali/VPU/PMIC; hardware secure-boot. This is strictly MORE fidelity than the shipped T0 emulator (which stubs apply entirely) and a genuine complement below Cuttlefish (T2).

Buildability + reuse (§11.4.74): highly buildable — two documented reproducible OSS recipes exist for A/B+U-Boot+QEMU, authored x86 with “only minimal tweaks to work with qemu-arm64”: RAUC (mature embedded A/B client, U-Boot-native, format=verity dm-verity slots, official QEMU example — RAUC examples <https://rauc.readthedocs.io/en/latest/examples.html> ; Pengutronix RAUC-on-QEMU <https://pengutronix.de/en/blog/2022-02-03-tutorial-evaluating-rauc-on-qemu-a-quick-setup-with-yocto.html>) and mind.be “Simulating an A/B update scheme using qemu” 2-part (Buildroot+U-Boot+RAUC, <https://mind.be/blog/2025/06/19/qemu-a-b-boot-x86-part-1.html> + part 2 = bootcount/altbootcmd rollback). Alternatives surveyed/not chosen: Mender (cloud-heavy), swupdate (own ecosystem), OSTree (file-based, wrong model for GPT-slot proof), casync (delta transport, orthogonal). **RAUC wins** (block-slot A/B + native U-Boot bootcount + verity + QEMU example = closest RK3588-A/B match, least new code).

Reuse decision: Catalogue–Check: reuse RAUC + U-Boot (upstream OSS, in-guest); extend vasic-digital/containers. New code = (a) guest-image build recipe, (b) thin pkg/vm A/B-disk + slot-state extension, (c) Helix OTA glue (agent ApplyPort → in-guest slot-writer). Reimplementing an A/B client = §11.4.74 violation.

Effort estimate (flagged estimate, not fact): first GREEN-on-clean-apply slice ~3–6 engineer-days (mostly Buildroot guest-image + U-Boot A/B env, copy-adaptable from cited recipes); corrupt-slot→auto-rollback proof ~+2–3 days. Cheapest real-A/B fidelity reachable on this Mac.

4. Q4 — containers integration + evidence plan

Reuse (present, FACT from containers/pkg/): pkg/vm (QEMU aarch64 virt+AAVMF, SSH/QMP clients, screendump, kvmAvailable() accel gating, –snapshot, distinct-port safety) — already the exact seam; pkg/boot+pkg/compose+pkg/health (on-demand boot/readiness, §11.4.76 invariant); pkg/emulator orphan-qemu reaping.

Extend upstream (PR to vasic-digital/containers, never copy): pkg/vm A/B-disk profile (attach 2nd virtio-blk GPT A/B disk; boot via U-Boot qemu_arm64 so bootcount env is live; expose slot-state read over existing SSH client; boot-attempt-counter probe); optional gated/reversible corrupt-slot injector (snapshot-backed via existing –snapshot, cannot harm host — §11.4.133 applied to virtual target).

Stays in helix_ota: agent ApplyPort→in-guest slot-writer glue (agent already depends only on ApplyPort, not update_engine — ota-android-agent README); per-tier test drivers; bank/Challenge wiring. Server unchanged (emulator registers/polls /api/v1 like T0 + a real board).

Evidence (§11.4.27/.107/.69/.108/.83): every PASS via ab_pass_with_evidence under docs/qa/<run-id>/, feature class boot_service. Runtime signatures = definition of done: inactive-slot block-hash delta; U-Boot env BOOT_ORDER flip; post-reboot findmnt /+/proc/cmdline=new slot; dmsetup status verity-active; bootcount/upgrade_available→0 good-marking; **headline rollback = altbootcmd fired + BOOT_ORDER reverted + root=old slot**. Test types: unit→integration(real QEMU+RAUC boot)→e2e/full-automation(apply→flip→reboot→verify→good AND corrupt→rollback)→chaos/stress §11.4.85 (kill-mid-apply resumable; power-cut-during-flip via QMP no-brick→rollback; N-iter determinism §11.4.50). No fakes beyond unit (§11.4.27). HelixQA bank challenge scoring on captured slot-state+rollback-trace. §1.1 mutation: strip corrupt-slot injection/rollback assertion → rollback test must FAIL. §11.4.116 JSONL verdict stream + snapshot, each verdict carrying its evidence path.

5. Honest fidelity ladder

Rung	Tech	This Mac now?	Proves	Gap
T0 (shipped)	ota-device-	YES	wire/server flow; agent decision	stubs entire apply

Rung	Tech	This Mac now?	Proves	Gap
A/B-virt (RECOMMENDED)	emulator (podman) QEMU virt+HVF Linux + U-Boot bootcount + RAUC dm-verity A/B	YES (HVF-accel)	REAL apply→slot-flip→dm-verity→auto-rollback on RK3588 CPU arch	NOT real Android update_engine/AVB; NOT RK3588 silicon
T1	Android AVD arm64 (HVF)	YES	real Android userspace + agent/bridge	NOT real A/B; GSI-A/B UNCONFIRMED
T2	Cuttlefish (cvd)	NO (/dev/kvm)	real Android update_engine A/B+AVB/dm-verity+rollback	host-gated §11.4.112
T3	real RK3588	NO (hardware)	+ vendor HAL + real U-Boot pkg + Mali + real partitions	hardware-gated §11.4.112

The recommended rung is the **ONLY** one exercising a genuine A/B apply+rollback mechanism on this Mac — honestly between T0 (no apply) and T2 (real Android apply, host-gated).

6. Recommended phased PWU plan (§11.4.58)

Build the QEMU virt+HVF Linux A/B emulator (Tfw-A/B tier, filling the unbuilt P-A/B gap in ROADMAP.md), reusing RAUC+U-Boot (in-guest) + extending containers/pkg/vm.

- **PWU-AB-1 (FIRST minimal buildable+verifiable slice):** Buildroot aarch64 guest (adapt mind.be/RAUC recipe) with 2-slot GPT + U-Boot bootcount A/B env; boot via extended pkg/vm on HVF; apply payload→slot B, flip active→B, reboot, assert root mounted=B. RED (RED_MODE=1, §11.4.115): “root==B after apply+flip+reboot” FAILs pre-flip. GREEN: PASSes after real apply; evidence = pre/post findmnt / + U-Boot env dump over SSH. **Proves a REAL slot switch on this Mac.**
- PWU-AB-2: dm-verity per slot (format=verity); dmsetup status verity-active; clean apply marks good.
- **PWU-AB-3 (headline):** corrupt inactive slot verity region; reboot; assert U-Boot altbootcmd fires + auto-rollback to previous good slot (the §11.4.108 rollback runtime signature).
- PWU-AB-4: wire ota-android-agent ApplyPort→in-guest slot-writer; full register→update-check→apply→reboot→verify→telemetry over ota-protocol vs live server.
- PWU-AB-5: chaos/stress §11.4.85 + HelixQA bank + §1.1 mutation.

Each PWU: pkg/vm extension PR'd upstream (§11.4.74); guest-image regen mechanism (§11.4.77, gitignored+rebuildable); evidence under docs/qa// (§11.4.83); independent review (§11.4.125/.142); §1.1 mutation. The FIRST slice is PWU-AB-1 — proves something real about A/B on this host with a RED→GREEN captured-evidence test, smallest scope, before dm-verity/rollback complexity.

7. §11.4.74 reuse decision (summary)

Component	Decision	Rationale
In-guest A/B client	reuse RAUC (upstream)	mature, U-Boot-native, format=verity, QEMU example
Slot select + rollback	reuse U-Boot bootcount/altbootcmd/upgrade_available	same engine real RK3588 U-Boot uses
dm-verity	reuse kernel dm-verity (RAUC verity slot)	identical kernel feature
VM lifecycle/boot/health	extend vasic-digital/containers pkg/vm,boot,health (§11.4.76)	seam already exists (aarch64 virt+HVF)
OTA glue (ApplyPort/protocol/server)	stays in helix_ota	OTA-specific, not generic plumbing
Guest image	build recipe (Buildroot), gitignored + §11.4.77 regen	build derivative, not versioned

8. What genuinely CANNOT be done on this Mac (§11.4.6/.112 — no bluff)

1. Real Android update_engine A/B + AVB/dm-verity + auto-rollback (Cuttlefish-grade): host-gated — needs Linux+/dev/kvm; M3 lacks nested-virt. Runs on Linux+KVM/M4+/GCE-nested-virt (the designed T2/P4 path). NOT structurally impossible.
2. RK3588 silicon fidelity (Mali-G610, VPU, Rockchip PMIC/clocks, vendor TPL/SPL/idbloader, secure-boot fuses): no QEMU RK3588 model exists; real-hardware-only (T3) or redroid-on-real-RK3588 (adjacent, no bootloader).
3. Stock-AVD real-A/B (GSI): UNCONFIRMED — needs the GSI-A/B empirical test; neither claimed nor relied upon.

The recommended A/B-virt rung attempts none of these — it proves the A/B mechanism (apply/slot/verity/rollback) honestly and defers silicon + real-Android-update_engine to T2 (Linux+KVM) and T3 (hardware), already designed in the fabric.

Sources verified 2026-06-11

- No QEMU RK3588 model; use generic virt: <https://forum.osdev.org/viewtopic.php?f=1&t=37324> ; <https://github.com/qemu/u-boot/blob/master/arch/arm/mach-rockchip/Kconfig>
- QEMU virt + U-Boot aarch64: <https://docs.u-boot.org/en/stable/board/emulation/qemu-arm.html>
- U-Boot A/B auto-rollback (bootcount/bootlimit/altbootcmd/upgrade_available): <https://docs.u-boot.org/en/latest/api/bootcount.html> ; <https://docs.mender.io/operating-system-updates-yocto-project/board-integration/bootloader-support/u-boot/manual-u-boot-integration>
- RAUC A/B + QEMU + format=verity dm-verity: <https://rauc.readthedocs.io/en/latest/examples.html> ; <https://pengutronix.de/en/blog/2022-02-03-tutorial-evaluating-rauc-on-qemu-a-quick-setup-with-yocto.html>
- A/B-on-QEMU reproducible recipe (“minimal tweaks for qemu-arm64”): <https://mind.be/blog/2025/06/19/qemu-a-b-boot-x86-part-1.html> ; <https://mind.be/blog/2025/06/20/qemu-a-b-boot-x86-part-2.html>
- Cuttlefish needs Linux+KVM (host-gated): <https://source.android.com/docs/setup/create/cuttlefish-use>
- GSI exists / from emulator images (AVD-A/B UNCONFIRMED): <https://source.android.com/docs/core/tests/vts/gsi> ; <https://developer.android.com/topic/generic-system-image/releases>
- Host facts: this host uname/sysctl/sw_vers 2026-06-11 + ../emulation_infra/REPORT.md §0

Negative findings / gaps (§11.4.99(B))

- No QEMU RK3588 board model in any upstream source surveyed — virt is the only realistic aarch64 target (absence across QEMU board lists + OSDev guidance).
- RAUC’s OFFICIAL full-system QEMU example is x86+GRUB, not arm64+U-Boot — but RAUC docs document the U-Boot bootcount A/B script and the mind.be recipe is U-Boot-based + “minimal tweaks for qemu-arm64”. arm64+U-Boot is reproducible but needs the recipe adaptation (bulk of PWU-AB-1 effort) — flagged, not hidden.
- AVB (Android Verified Boot vbmeta) vs plain dm-verity: recommended slice uses plain kernel dm-verity (sufficient to prove verify+rollback). Adding AVB vbmeta for higher Android-fidelity is UNCONFIRMED and deferred (not required to prove the A/B mechanism; better suited to T2 Cuttlefish).